DiaTwin

Diabetes Digital Twin

A real-time pharmacodynamic platform for insulin-requiring diabetes, turning passive logging into predictive self-care.

IOB Tracking

MOB Modelling

Dose Advisory



8.7/10

Clinical Utility Score

50K+

Meal-outcome pairs validated

Advisory use only — does not replace clinical judgement

THE PROBLEM

Managing T1D is a 24/7 balancing act



The Precision Gap

Doctor-set therapy plans are complex to apply daily. Patients lack tools to translate clinical parameters into real-time decisions.



Data Without Insight

CGM and finger-prick devices show current glucose levels but offer no guidance on how much insulin to take or when.



Risk of Harm

Without IOB/MOB awareness, insulin stacking causes dangerous hypoglycemia. High cognitive burden leads to dosing errors.

WHAT IS DIATWIN?

A pharmacodynamic digital twin for T1D

DiaTwin builds a real-time model of each patient's glucose state tracking how insulin and carbohydrates behave physiologically to answer three critical questions every minute:

- **What is my BG right now?**
- **Where will it go in the next few hours?**
- **Is it safe to eat or dose?**

IOB Insulin On Board

Insulin still pharmacologically active, calculated via trapezoidal integration

MOB Meal On Board

Carbohydrates still being absorbed, modelled via gamma-distribution kinetics

Net Net BG Effect

Meal impact minus insulin impact, positive = rising, negative = falling

Advisory use only — does not replace clinical judgement

THE ENGINE

Pharmacodynamics Engine

Meal Absorption (MOB)

Gamma-distribution curve modelling:

- Delayed gastric emptying onset
- Peak absorption rise
- Gradual metabolic tail
- 5 absorption speeds: very slow → very fast
- Personalised absorption modifier per patient

Insulin Activity (IOB)

Two curve families — S3 Clinical Guidelines:

Peaked Insulins

- Rapid, short, intermediate acting
- Gamma distribution with type-specific params

Peakless Insulins

- Long-acting, ultra-long acting
- 4-phase model, capped at 75% max activity

FOR PATIENTS

Predictive Self-Care



Real-Time IOB & MOB

See exactly how much insulin and meal carbs are still active, eliminating guesswork before a correction dose.



Advisory Bolus Calculator

Dose suggestions accounting for current BG, active IOB, planned meal, activity level, and time-of-day resistance.



Pre-Bolus Timing

Recommends optimal pre-meal injection window (0–20 min) based on the glycaemic index of the logged meal.



Structured Meal Logging

Extensive food database including local cuisine categories reducing carb-counting burden and improving adherence.



BG Rise Prediction

MOB is converted into an estimated future glucose rise letting patients anticipate hyperglycaemia before it occurs.



Activity Adjustment

Physical activity logging automatically adjusts dose suggestions supporting safer exercise management.

FOR PHYSICIANS

Evidence-Based Remote Monitoring

Remote Patient Monitoring

Authorised physicians access a dedicated dashboard: BG history, meal logs, insulin records, and activity data for informed pre-consultation review.

Glycaemic Response Analysis

Meal-glucose correlation charts reveal postprandial response patterns across days or weeks objective evidence for ICR or ISF titration.

Clinical Parameter Review

All patient-specific constants (ICR, CF, target glucose, absorption modifiers) are visible in a structured view.

Pattern-Based Titration

Time-in-range statistics, SMBG analysis, and glycaemic variability metrics (MAGE, CV%) provide a structured evidence base for dose conversations.

Consent-Gated Access

Bilateral authorisation model ensures patient data sovereignty. Physicians see only consented patients — GDPR-aligned clinical data governance.

Multi-Patient Overview

Full panel of authorised patients, each searchable by name, enabling efficient review workflows in specialist clinics or GP settings.

THE DASHBOARD

At-a-Glance Metabolic Control

ActiveEffectsDisplay

Primary home-screen widget — refreshes every 60 seconds synced to wall-clock :00 boundary

Last Meter Reading	Most recent fingerstick with time-since badge (e.g. '14 min ago')
Insulin Card (IOB)	Further drop still to come
Meal Card (MOB)	Hidden rise still coming
Current Est. BG	Baseline + cumulative net colour-coded green/orange/red
Projected Final BG	Where BG will land if nothing new is added

Additional Visualizations

EffectsVisualization

IOB + MOB evolution over next few hours

BloodSugarVisualization

Full BG history vs. model estimates

MealVisualization

Active meals, absorption rates, pending BG rise

InsulinVisualization

Dose history, active insulin, stacking risk

ARCHITECTURE

Full-Stack Cross-Platform System

Mobile (React Native / Expo)

- iOS + Android + Web
- TypeScript components
- Offline-capable with sync
- Libre CGM integration

Frontend (React Web)

- JS + TS components
- Real-time visualization
- Doctor & Patient dashboards
- Import/Export support

Backend (Python / Flask)

- MongoDB database
- Pharmacodynamics engine
- JWT authentication
- REST API endpoints

Single source of truth: pharmacodynamic constants shared between Python backend and TypeScript frontend — zero drift

MARKET

A Growing, Underserved Market

+5M

Insulin-pen users lacking structured dosing guidance

8.7%

CAGR for global diabetes digital management market

~26K

Conservative 2% German market penetration within 3 years

€100/yr

DiaTwin cost vs €3–5K/yr for CGM hardware

Competitive Landscape

Feature	DiaTwin	CGM Only	Basic Apps	Insulin Pumps
Intelligent Dosing	✓ IOB, MOB, Projection	✗	✗	✓ Closed-loop only
Works with Insulin Pens	✓	✓	✓	✗
Annual Cost	~€100	€3,000–5,000	Low	€10,000–16,000

VALIDATION & TEAM

Built on Clinical Evidence

Grade 1.6 'Excellent'

Master's Thesis

Core algorithms validated at DIT

8.7 / 10

Clinical Utility

Scored by certified diabetologist

50,000+ pairs

Dataset

Real clinical meal-outcome data

TRL 4–5

Readiness

Web + mobile prototypes operational

ROADMAP

Path to CE IIa Certification & DiGA

Now

TRL 4–5

- Core algorithms validated
- Web + mobile prototypes
- Clinical utility 8.7/10

Phase I

EXIST Funding

- Team expansion
- Full-stack dev
- DiGA strategy co-founder

Phase II

TRL 6–7

- Clinical trial N=100, 6 months
- Safety/efficacy validation
- Physician workflow integration

Phase III

Regulatory

- CE Class IIa certification
- DiGA pathway approval
- International expansion

WHO WE SERVE

Two Target Patient Groups

Continuous CGM Users

Already wearing a CGM sensor — but missing structured dosing guidance to act on their data.

What DiaTwin adds:

- Net Effect analysis on live glucose trends
- 4-hour BG projection (IOB + MOB overlay)
- Intelligent bolus advice beyond trend arrows
- Reduces unnecessary corrective doses
- Saves €3,000–5,000/year vs. CGM hardware alone

CGM Companion — keeps precision, slashes cost

vs

SMBG / Finger-Prick Users

Global · Largest underserved group

3–4 daily finger-sticks — significant gaps between readings where glucose direction is unknown.

What DiaTwin provides:

- Intelligent Glucose Estimation between readings
- MOB modelling (5 meal types, portion-based)
- IOB tracking across overlapping doses
- Future glucose projection without CGM
- Trend insight at a fraction of CGM cost (~€100/yr)

CGM-level insight — without the sensor

PHYSICIAN CONTROL

All Parameters Set & Managed by Doctors

DiaTwin does not allow patients to modify clinical parameters. Every threshold, ratio, and limit is prescribed and adjusted exclusively by the treating physician — ensuring safety, clinical accountability, and regulatory compliance.

Insulin-to-Carb Ratio (ICR)

Physician only

Defines how many grams of carbohydrate one unit of insulin covers. Personalised per patient and time-of-day.

Correction / Sensitivity Factor

Physician only

How much 1 unit of insulin lowers blood glucose. Used to calculate correction boluses against a target BG.

Target Blood Glucose

Physician only

The individual BG goal for each patient — used to calculate the correction delta and colour-code estimates.

Safety Limits & Dose Caps

Physician only

Maximum allowed bolus per dose and per day. Hard stops prevent the system from suggesting unsafe corrections.

Absorption Modifiers

Physician only

Patient-specific multiplier that scales the gamma-distribution absorption curves for meals and insulin activity.

Basal Rate Equivalent

Physician only

Long-acting insulin baseline parameters — onset, duration, and activity profile used in IOB calculations.

Patients interact only with logging and viewing functions. Parameter access is gated behind physician authentication.

DiaTwin

Diabetes Digital Twin

From reactive monitoring
to predictive self-care.

- Pharmacodynamic precision for every patient, every dose
- Real-time IOB + MOB — glucose predicted, not just measured
- Physician-supervised, patient-centred, GDPR-compliant

Advisory use only — does not replace clinical judgement